# CAMBRIDGE ASSESSMENT INTERNATIONAL EDUCATION

**Cambridge IGCSE™ & O Level Computer Science (0478 / 2210)**

**OFFICIAL MARKING SCHEME & COMPLETE WORKED SOLUTIONS**

**TOPICS 1.1 & 1.2: Data Representation (2023 – 2026 Series)** **Maximum Mark: 60**

## SECTION A: NUMBER SYSTEMS (1.1)

### Question 1 [Total: 6 marks]

**(a) Binary to Denary:** [1 mark]

01011010 = 64 + 16 + 8 + 2 = **90** [1]

**(b) Two's Complement of −45:** [2 marks]

• Step 1: Represent +45 in 8-bit binary: `00101101` [1]

• Step 2: Invert bits (1's comp): `11010010` → Add 1: `11010011` [1]

*(Alternative: Working from right, keep bits up to first 1 then invert remainder → 11010011)*

**(c) Binary Addition & Overflow:** [3 marks]

```
  0 1 0 1 1 0 1 0  (90)
+ 0 0 1 1 0 1 1 1  (55)
-----------------
  1 0 0 1 0 0 0 1  (145)
```

[1 mark for correct binary sum]

• **Overflow Error: No overflow error has occurred** [1]

• **Reason:** The result (145 / `10010001`) fits within the 8-bit register range (0 to 255) / there is no carry out from the 8th most significant bit (MSB) [1].

### Question 2 [Total: 6 marks]

**(a) Hexadecimal 3B Conversion:** [2 marks]

• Binary: 3 = `0011`, B (11) = `1011` → **00111011** [1]

• Denary: (3 × 16) + 11 = 48 + 11 = **59** [1]

**(b) Reasons for using Hexadecimal:** [2 marks]

*[1 mark per valid point, max 2]*

• Shorter and more compact representation than binary (takes up less screen/memory space).

• Much easier for humans to read, write, and remember than long binary strings.

• Reduces human transcription errors during programming and debugging.

• Quick and easy to convert directly to and from binary (1 hex digit = 4 bits / 1 nibble).

**(c) Uses of Hexadecimal:** [2 marks]

*[1 mark per valid example, max 2]*

• MAC addresses (e.g. `00:1A:2B:3C:4D:5E`)

• IPv6 addresses

• HTML / CSS / RGB Colour Codes (e.g. `#FF5733`)

• Memory Dumps / Error Codes in debugging

• Assembly language / Machine code opcode display

### Question 3 [Total: 5 marks]

**(a) Logical Left Shift of 2 Places:** [2 marks]

• Original: `0 0 1 1 0 1 0 0` (52)
• Shift 2 places left: **1 1 0 1 0 0 0 0** [1]
• New Denary: 128 + 64 + 16 = **208** [1]

**(b) Arithmetic Effect & Overflow:** [3 marks]

• **Arithmetic Effect:** Each 1-place left shift multiplies the denary number by 2 (a 2-place shift multiplies by 4 → 52 × 4 = 208) [1.5]
• **Effect of bits shifting out of MSB:** If a '1' bit is shifted out of the left-most bit (MSB), an **overflow error** occurs, causing the number to lose precision / become mathematically incorrect [1.5].

## SECTION B: TEXT REPRESENTATION (1.2.1)

### Question 4 [Total: 6 marks]

**(a) ASCII vs Unicode:** [3 marks]

• **ASCII:** Uses 7 bits (128 characters) or 8 bits / Extended ASCII (256 characters); only supports English letters, standard numbers, and symbols [1.5].
• **Unicode:** Uses 16 or 32 bits (e.g. UTF-8 / UTF-16) capable of representing over 1 million characters; supports all world languages (e.g. Arabic, Bengali, Chinese), mathematical symbols, and emojis [1.5].

**(b) Text File Size Calculation:** [3 marks]

• Total characters = 2,048 characters
• Bits = 2,048 × 8 bits = 16,384 bits
• Bytes = 16,384 ÷ 8 = 2,048 Bytes [1]
• In Kibibytes (KiB) = 2,048 ÷ 1,024 = **2 KiB** [2 marks for correct final answer with working]

## SECTION C: SOUND REPRESENTATION (1.2.2)

### Question 5 [Total: 7 marks]

**(a) Analogue to Digital Sound Conversion:** [3 marks]

• Sound waves are continuous analogue signals picked up by a microphone [1].
• The analogue wave is sampled at regular time intervals using an ADC (Analogue to Digital Converter) [1].
• The amplitude of each sample is measured and converted into a binary number based on the sample resolution [1].

**(b) Definitions:** [2 marks]

• **Sample Rate:** The number of sound samples captured per second, measured in Hertz (Hz) / kHz [1].
• **Sample Resolution (Bit Depth):** The number of bits allocated to store the amplitude/value of each sample [1].

**(c) Effect of Increasing Sample Rate:** [2 marks]

• **File Size:** Increases significantly (more data points recorded per second) [1].
• **Sound Quality:** Improves / sound is clearer and more faithful to the original wave [1].

### Question 6 [Total: 4 marks]

**Sound File Size Calculation:** [4 marks]

• **Formula:** File Size = Sample Rate (Hz) × Sample Resolution (bits) × Time (s) × Channels [1]
• **Calculation in bits:** 44,100 × 16 × 10 × 1 = 7,056,000 bits [1]
• **In Bytes:** 7,056,000 ÷ 8 = 882,000 Bytes [1]
• **In Megabytes (MB):** 882,000 ÷ 1,000,000 = **0.882 MB** (or 882,000 ÷ 1,048,576 = **0.841 MiB**) [1]

## SECTION D: IMAGE REPRESENTATION (1.2.3)

### Question 7 [Total: 7 marks]

**(a) Terms in Bitmap Images:** [3 marks]

• **Pixel:** The smallest individual picture element of a digital bitmap image that can be assigned a specific colour [1].
• **Image Resolution:** The total number of pixels in an image, expressed as width × height (or DPI - dots per inch) [1].
• **Colour Depth:** The number of bits used to represent the colour of each single pixel (e.g. 24-bit True Colour allows $2^{24} = 16,777,216$ colours) [1].

**(b) Image File Size Calculation:** [4 marks]

• **Formula:** Total Pixels × Colour Depth = Width × Height × Colour Depth [1]
• **Calculation in bits:** 1000 × 800 × 24 = 19,200,000 bits [1]
• **In Bytes:** 19,200,000 ÷ 8 = 2,400,000 Bytes [1]
• **In Mebibytes (MiB):** 2,400,000 ÷ (1024 × 1024) = 2,400,000 ÷ 1,048,576 ≈ **2.29 MiB** (or 2.4 MB in decimal) [1]

## SECTION E: DATA COMPRESSION (1.2.4)

### Question 8 [Total: 6 marks]

**(a) Lossy vs Lossless Compression:** [4 marks]

• **Lossy Compression:** Permanently removes unnecessary, redundant, or imperceptible data (e.g. sounds outside human hearing range, subtle colour differences). The original file cannot be recreated exactly, but achieves a very high compression ratio [1.5].
*Example format:* **JPEG, MP3, MP4** [0.5]
• **Lossless Compression:** Compresses file size by identifying and replacing patterns without deleting any data. When decompressed, the original file is reconstructed bit-for-bit exactly with zero loss of quality [1.5].
*Example format:* **PNG, GIF, FLAC, ZIP, SVG** [0.5]

**(b) Importance of Compression:** [2 marks]

• Reduces download and transmission time / faster webpage loading [1].
• Consumes less network bandwidth and reduces required storage space on disk / cloud [1].

## Question 9 [Total: 4 marks]

**(a) Run-Length Encoding (RLE):** **[2 marks]**

• RLE is a lossless compression algorithm that detects consecutive repeated values (runs) in a data stream [1].

• It replaces the repeated values with a pair consisting of the **count/frequency** followed by the **value/code** [1].

**(b) RLE Compression Example:** **[2 marks]**

• Original: `W W W W W  B B B  W W  G G G G` (5 W, 3 B, 2 W, 4 G)

• **RLE Stored Format: 5W 3B 2W 4G** (or `05 W 03 B 02 W 04 G`) [2 marks for complete correct encoding]